COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S3
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PRELIM SKILL EXAM **The Palindrome Number Problem**

1. PROCEDURE OUTPUT

Problem Title: The Palindrome Number Problem

Problem Definition:

A palindrome number is a number that remains the same when digits are reversed. For example, the number 12321 is a palindrome number, but 1451 is not a palindrome number. Given a positive integer, the task is to check whether the number is palindrome or not. We want to use data structures to check whether a given string of input conforms to the given description of a palindrome. The user is then prompted whether the given input is a palindrome or not.

Your task is to use two data structures for your solution:

- Stack
- Queue

But, they must have different internal representations, either:

- Array-based
- Linked List

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Palindrome?	Output (Console)	Remark
10201	Palindrome Number Checker: Enter a integer: 10201 10201 is a palindrome number.	Success
1003003001	Palindrome Number Checker: Enter a integer: 1003003001 1003003001 is a palindrome number.	Success

88	Palindrome Number Checker: Enter a integer: 88 88 is a palindrome number.	Success
0	Palindrome Number Checker: Enter a integer: 0 0 is a palindrome number.	Success
202	Palindrome Number Checker: Enter a integer: 202 202 is a palindrome number.	Success

Analysis:

This program determines whether a positive integer is a palindrome by using two data structures: an array-based Stack and a linked-list-based Queue. Each digit entered by the user is stored in both the Stack and Queue. Since the Stack follows LIFO, it returns the digits in reverse order, while the Queue follows FIFO, returns the digits in their original order. It then compares the characters removed from the Stack and Queue one by one. If all corresponding characters are equal, the number is a palindrome. If any pair is different, the program immediately determines that the number is not a palindrome. The Stack has a maximum capacity of 100 characters, while the Queue dynamically creates nodes as needed. The Queue also has a destructor that properly deletes the nodes when the Queue is no longer needed, preventing memory leaks. It demonstrates how Stack and Queue data structures can work together to solve a palindrome- checking problem. The time complexity is O(n) because each digit is processed once, while the space complexity is O(n) because the digits are stored in both data structures.
The isPalindrome() removed from the Stack and Queue. If all values match, the number is a palindrome; otherwise, it is not. The main() function accepts the user's input and displays the result.

Pseudo code:

```
START

FUNCTION isPalindrome(number)

    CREATE an empty Stack
    CREATE an empty Queue

    FOR each character digit in number
        PUSH digit into Stack
        ENQUEUE digit into Queue
    END FOR

    WHILE Stack is not empty

        stackValue ← POP from Stack
        queueValue ← DEQUEUE from Queue

        IF stackValue is not equal to queueValue
            RETURN false
        END IF

    END WHILE

    RETURN true

END FUNCTION

MAIN
    DISPLAY "PALINDROME NUMBER CHECKER"
    INPUT number
    IF isPalindrome(number) is true
        DISPLAY number + " is a palindrome number."
    ELSE
        DISPLAY number + " is not a palindrome number."
    END IF

END MAIN
```

Code:

Main.cpp

```cpp
#include <iostream>
#include <string>
#include "Stack.h"
#include "Queue.h"

bool isPalindrome(const std::string& number)
{
    Stack stack;
    Queue queue;

    for (char digit : number)
    {
        stack.push(digit);
        queue.enqueue(digit);
    }

    while (!stack.isEmpty())
    {
        char stackValue = stack.pop();
        char queueValue = queue.dequeue();

        if (stackValue != queueValue)
        {
            return false;
        }
    }

    return true;
}

int main()
{
    std::string number;
    std::cout << "Palindrome Number Checker:\n";
    std::cout << "Enter a integer: ";
    std::cin >> number;

    if (isPalindrome(number))
    {
        std::cout << number << " is a palindrome number.\n";
    }
    else
    {
        std::cout << number << " is not a palindrome number.\n";
    }

    return 0;
}
```

Queue.h

```cpp
#ifndef QUEUE_H
#define QUEUE_H

struct Node
{
    char data;
    Node* next;

    Node(char value)
    {
        data = value;
        next = nullptr;
    }
};

class Queue
{
private:
    Node* front;
    Node* rear;

public:
    Queue()
    {
        front = nullptr;
        rear = nullptr;
    }

    ~Queue()
    {
        while (!isEmpty())
        {
            dequeue();
        }
    }

    bool isEmpty() const
    {
        return front == nullptr;
    }

    void enqueue(char value)
    {
        Node* newNode = new Node(value);
        if (isEmpty())
        {
```

```
        if (isEmpty())
        {
            front = newNode;
            rear = newNode;
        }
        else
        {
            rear->next = newNode;
            rear = newNode;
        }
    }

    char dequeue()
    {
        if (isEmpty())
        {
            return '\0';
        }
        Node* temp = front;
        char value = temp->data;
        front = front->next;

        if (front == nullptr)
        {
            rear = nullptr;
        }
        delete temp;

        return value;
    }
};

#endif
```

Stack.h

```
#ifndef STACK_H
#define STACK_H

class Stack
{
private:
    static const int MAX = 100;
    char data[MAX];
    int top;

public:
    Stack()
    {
        top = -1;
    }

    bool isEmpty() const
    {
        return top == -1;
    }

    bool isFull() const
    {
        return top == MAX - 1;
    }

    void push(char value)
    {
        if (!isFull())
        {
            data[++top] = value;
        }
    }

    char pop()
    {
        if (!isEmpty())
        {
            return data[top--];
        }

        return '\0';
    }
};

#endif
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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